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Antibody capture process based on magnetic beads from very high cell density suspension

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Abstract

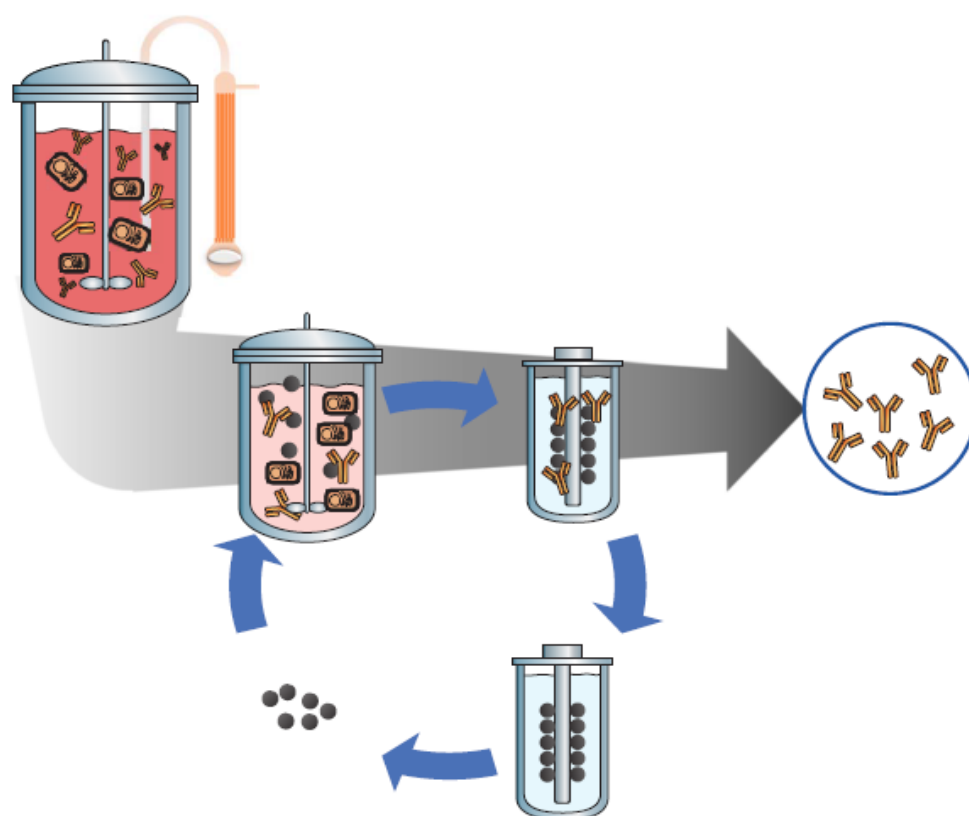
Cell clarification represents a major challenge for the intensification through very high cell density in the production of biopharmaceuticals such as monoclonal antibodies (mAb). The present report proposes a solution to this challenge in a streamlined process where cell clarification and mAb capture are performed in a single step using magnetic beads coupled with protein A. Capture of mAb from non-clarified CHO cell suspension showed promising results, however it has not been demonstrated that it can handle the challenge of very high cell density as observed in

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intensified fed-batch cultures. The performances of magnetic bead-based mAb capture on non-clarified cell suspension from intensified fed-batch culture were studied. Capture from a culture at density larger than 100×10^6 cells/mL provided an adsorption efficiency of 99 % and an overall yield of 93 % with a logarithmic host cell protein (HCP) clearance of ≈ 2 -3 and a resulting HCP concentration $\leq \approx 5$ ppm. These results show that direct capture from very high cell density cell suspension is possible without prior processing. This technology, which brings significant benefits in terms of operational cost reduction and performance improvements such as low HCP, can be a powerful tool alleviating the challenge of process intensification.

Graphical Abstract



The authors present a magnetic bead-based purification for the direct capture of monoclonal antibodies from very high cell density ($> 100 \times 10^6$ cell/mL)

intensified fed-batch CHO cell culture. Direct capture from non-clarified cell suspension showed excellent efficacy regarding absorption, elution, and very low final HCP content, despite a very high cell density. Compared to legacy cell clarifications following mAb capture, this process represents significant savings in operational time, consumables, and workload, while providing very gentle conditions for the cells.

1 Introduction

Intensification of cell culture processes based on medium renewal has gained vibrant interest this last decade. Different approaches such as increased inoculation cell density using perfusion mode in N-1, steady-state perfusion operation or hybrid production process gathering the advantages of fed-batch and perfusion modes have notably increased the volumetric mAb productivity (Mellahi et al. 2019; Pinto et al. 2020; Xu et al. 2020a; Xu et al. 2020b; Xu et al. 2017). The present report focusses on this latter concept; intensified fed-batch, also called concentrated fed-batch, is a process in which the culture medium is renewed using an ultra-filter in the cell separation device (Clincke et al. 2013b). It enables very high cell concentration and accumulation of the product of interest in the bioreactor, e.g. titers exceeding 25 g/L and cell densities over 200×10^6 cells/mL (Clincke et al. 2013a; Xu et al. 2017; Yang et al. 2016). The very high cell mass of this type of process or steady-state perfusion represents a severe challenge for the cell clarification step (Popova et al. 2016). Generally speaking, this step is costly due to the equipment and/or consumables (Popova et al. 2016), time intensive, and can potential cause increased host cell protein (HCP) levels from cell lysis (Hogwood et al. 2013; Pohlscheidt et al. 2013). At very high cell density, centrifugation and filtration methods can reach their

limitation of removing solid particles (Singh et al. 2013). Centrifugation techniques are able to deal to a great extent with solid particles, but even hermetically-sealed centrifuges can generate significant cell disruption that can decrease clarification efficiency and cause product degradation (Singh et al. 2013). Additionally, the feed-stream can be challenging for the depth filtration step carried out after centrifugation to remove solids, leading to filter clogging or poor utilization of the filter (Popova et al. 2016; Singh et al. 2013). In order to improve the cell clarification step, pretreatment of the feed stream with flocculation agents and low pH can be used. Those pretreatments can lead to a better clarification but have the drawbacks of adding new components in the culture or unsuitable low pH for the following capture step (Hadpe et al. 2020; Liu et al. 2010; Popova et al. 2016). The cell clarification of high cell density culture remains thus a bottleneck in manufacturing.

Following the cell clarification, the capture step using protein A column chromatography is the gold standard for mAb purification due to its high affinity for these proteins (Branco et al. 2012; Ghose et al. 2005; Kelley 2009). This step represents however another bottleneck. The protein A column chromatography performance can hardly match the production from the intensified cell cultures (Käppler and Posten 2008; Shukla and Thömmes 2010; Swartz et al. 2018). The major drawbacks of a column chromatography capture step are the long loading and wash phases (Holschuh and Schwämmle 2005) and the need to obtain cell-free supernatant (Kelley 2009).

Separation using magnetic beads, 'magnetic separation', is widely known (Yavuz et al. 2009) and has been used as analytical tool for several decades (Dunlop et al. 1984; Dunnill and Lilly 1974; Ebeler et al. 2019; Whitesides et al. 1983). Recently, the interest of using magnetic separation in purification process for

manufacturing has increased as an alternative to traditional purification techniques (Brechmann et al. 2019; Ebeler et al. 2018; Ebeler et al. 2019; Franzreb et al. 2006; Holschuh and Schwämmle 2005). This interest is motivated by favorable properties such as rapidity and capacity to process whole cell broth in a gentle manner for the cells (Ebeler et al. 2018; Holschuh and Schwämmle 2005; Hubbuch and Thomas 2002; Safarik and Safarikova 2004). Magnetic separation typically requires functionalized magnetic beads and a magnetic filter (Elmizadeh et al. 2013; Gañán et al. 2013; He et al. 2014; Yavuz et al. 2009). Functionalizing the magnetic beads with protein A combines the benefits of magnetic separation with the high affinity of these ligands for mAbs. This combination enables the possibility of a direct mAb capture from non-clarified cell broth and has been demonstrated at pilot scale up to 14×10^6 cells/mL in a previous report (Brechmann et al. 2019). Capture step by magnetic separation has been applied to different crude liquids (Ebeler et al. 2018; Ebeler et al. 2019), however, to our knowledge, no report of the challenging task of mAb capture from non-clarified high cell density culture obtained in intensified fed-batch cultures, has been published. A single-step process of magnetic separation integrating cell separation and capture of the product of interest directly applied to high cell density broth has the potential to provide major savings of operation time, consumables, and workload, while providing gentle conditions for the cells.

The purpose of the present work was to provide a proof-of-concept demonstrating magnetic bead mAb separation of non-clarified cell suspension at very high cell density. In particular, the objective was to apply this technology downwards of intensified fed-batch process using an UF-HF filter-based perfusion with retention of molecules above a given cut off. This latter process is known to be highly challenging for the cell clarification, and represents an excellent challenge to

demonstrate the potential of magnetic separation, including a significant simplification of the operational steps. Here we aimed at maintaining a high product yield over the mAb capture from non-clarified high cell density suspension, with purity comparable or improved compared to legacy approach of cell clarification followed by column chromatography.

2 Materials and Methods

2.1 Cells

For all the cultures, a stable CHO-M cell line expressing humanized IgG1 antibody, kindly provided by Selexis (Switzerland) was used. Cells were thawed and routinely passaged in BalanCD CHO Growth A medium, kindly provided by FUJIFILM Irvine Scientific (Santa Ana, CA, USA), supplemented with 4 mM glutamine and cultured at 37°C, 5 % CO₂ atmosphere and 120 rpm prior to inoculation of the perfusion culture.

2.2 Intensified fed-batch cultures

Intensified fed-batch cultures were performed in glass stirred tank DASbox bioreactors (Eppendorf, Hamburg, Germany) in 200 mL working volume, according to the method for perfusion operation described in Schwarz et al. 2020, except for the hollow fiber cartridges, which were ultra-filters. Three cultures, cult_01, cult_02, cult_03, were performed using alternating tangential flow XCell ATF2 (Repligen, Waltham, MA, USA) cell separation device mounted with a 50 kDa cutoff ultra-filtration hollow fiber filter (UF HF). For cult_01 and cult_02, the filter was F2:RF50PS with 0.14 m² membrane area (Repligen, Waltham, MA, USA). Notice that cult_01 was a preliminary culture to test the technology and was performed with a filter, which had already been used in a previous run and was sanitized. A filter

RTPUFP-50-C-4MS with 0.065 m² membrane area (Ready to process, Cytiva, Uppsala, Sweden) was used for cult_03. The perfusion using the 50 kDa cutoff UF HF provided exchange of all the components of low molecular weight while the larger molecules were retained in the bioreactor by the filter, resulting in accumulation of the antibodies. The cultures were perfused using feed medium composed of BalanCD CHO Growth A supplemented with 5 % BalanCD CHO Feed 4 (FUJIFILM Irvine Scientific, Santa Ana, CA, USA), 100 mg/L streptomycin and 60 mg/L penicillin G (Sigma Aldrich, St. Louis, MI, USA). Runs cult_01 and 02 were inoculated at 5×10^6 cells/mL and the perfusion was started the same day, while cult_03 was inoculated at cell density 1×10^6 cells/mL, and the perfusion started at day 2. The perfusion rate was initiated at 0.5 reactor volume/day and progressively increased with the cell density until day 10, after which the cell specific perfusion rate (CSPR) was set to 50 and 20 pL/cell/day for run_02 and run_03 respectively. Glucose and glutamine were supplied according to cell need. The set points of the temperature, dissolved oxygen concentration and pH were 37°C, 40 % and 7, respectively. The cultures were interrupted to proceed with the capture step at a given time point or when the UF HF clogged, causing dysfunction of the perfusion operation.

2.3 Purification of mAb using magnetic protein A agarose beads

The magnetic beads were MAGicBeads mAb (MAGic Bioprocessing, Uppsala, Sweden), described in a previous publication (Brechmann et al. 2019). Notice that the previously reported LOABeads PrtA (Brechmann et al. 2019) have been rebranded as MAGicBeads mAb beads. The magnetic separator for the capture of harvest from bioreactor runs was MAGic SEP 500 aimed for 500 mL size (MAGic Bioprocessing, Uppsala, Sweden).

The purification runs, run_01, run_02 and run_03, performed with crude cell suspension obtained from the intensified fed-batch runs cult_01, cult_02 and cult_03 were performed similarly with details listed in Table 1. Due to potential filter clogging at the harvest day, the final culture volume diverged somewhat from the working volume of 200 mL applied during the whole run. In order to apply similar conditions for run_02 and run_03, 200 mL of cell suspension were used for the purification, independently of the final volume at harvest day in the bioreactor.

The volumes of settled MAGicBeads mAb beads used for the purification were selected based on our previous work performed with the same beads and mAb (Brechmann et al. 2019). In short, in this report, the binding capacity of the beads per mass IgG had been determined for IgG concentrations between 1 and 8 g/L during a contact time of one hour. The bead volumes for the present runs were selected based on this information taking into account the respective IgG concentrations. For IgG concentration > 8 g/L, the binding at 8 g IgG/L was taken into account since the binding pattern was asymptotical at this concentration. The MAGicBeads mAb beads were equilibrated with PBS pH 7.4 and added to 200 mL of non-clarified cell suspension in a 500 mL Duran bottle (DKW Life Science GmbH, Wertheim/Main, Germany) for the loading phase. Continuous horizontal shaking with a table top shaker (Heidolph Unimax 2010, Heidolph instruments GmbH & CO. KG, Schwabach, Germany) was carried out at 150 to 180 rpm to prevent bead settling. After 120 min the beads were isolated using the magnetic field of the MAGic SEP 500. The unbound solid fraction was subsequently displaced with several PBS washes; the first PBS wash eliminated the cells and cell debris, while the subsequent PBS washes removed the unspecific binding of molecules such as proteins. Three PBS washes were used for cell density $\leq 20 \times 10^6$ cells/mL and 5 washes for higher

cell densities. The beads were then transferred into a semi-pack column of the bead volume to carry out the elution (100 mM citrate, pH 2.8). This was followed by immediate neutralization of the concentrated mAb by adding 2 M Tris-HCL, pH 9.0.

Furthermore, to monitor the adsorption, 1 mL samples were taken 5, 10, 15, 30, 60, and 120 minutes after contact of the beads with the cell suspension, centrifuged and stored at -80°C for later analysis.

2.4 Process analytics

Daily samples were taken from the culture to monitor the process parameters; viable cell density, total cell density, cell diameter and viability were measured with NORMA XS from iPRASENSE (Montpellier, France). Culture metabolites (glucose, glutamine, lactate and ammonium) and lactate dehydrogenase level were measured with a Cedex Bio Analyzer (Roche, Basel, Switzerland). The osmolality was measured on a Bioprofile Flex analyzer (Nova Biomedical, Waltham, MA, USA). The concentration of mAb in all the culture and purification samples was determined with the IgG assay from the Cedex Bio Analyzer (Roche, Basel, Switzerland).

2.5 Size Exclusion Chromatography

The detection of monomers, oligomers and low molecular fragments was performed with size exclusion chromatography (SEC). For the size separation a TSK Gel G3000SWXL, 7,8 mm × 300 mm size-exclusion column (Tosoh Bioscience, Tokyo, Japan) mounted to a Waters 2695 HPLC unit (Milford, MA, USA) was used, connected to a Waters 2996 UV detector (Milford, MA, USA) for absorbance measurement at 280 nm. Prior to SEC analysis, eluted mAb sampled were diluted while no dilution was needed for the cell culture supernatant samples. 20 µl sample were injected into the column and eluted for 70 min at a constant flow rate of 0.3

mL/min. The system used 100 mM sodium sulfate and 25 mM Na₂HPO₄ at pH 7.0 as mobile phase. Monomer percentage was calculated from the integral of the non-normalized peaks in relation to the remaining oligomer and fragmentation peaks.

2.6 Charge variant analysis

Charge variations of eluted mAbs were analyzed with a MAbPac™ SCX-10 10 µl column connected to a 2695 HPLC separation module (Waters, Milford, MA, USA) and a 2996 photodiode array detector (Waters, Milford, MA, USA). For column equilibration 95:5 % 20 mM MES, pH 5.6, 60 mM NaCl and 20 mM MES, pH 5.6, 300 mM NaCl was used as the mobile phase. The charge variants were eluted with a gradient ranging from 95:5 % to 45:55 % of 20 mM MES, pH 5.6, 60 mM NaCl and 20 mM MES, pH 5.6, 300 mM NaCl for 62 min. Samples were detected at 280 nm wavelength.

2.7 N-glycosylation analysis

N-glycosylation of the eluted mAbs was analyzed with GlycoWorks™ RapiFluor-MS™ N-Glycan Kit (Waters, Milford, MA, USA). Samples were treated by cleaving the N-glycosylation from the mAb, followed by fluorophore and basic tertiary amine tag labeling. After this pre-treatment, the samples were analyzed using an ACQUITY UPLC H-Class PLUS system (Waters, Milford, MA, USA) with an ACQUITY UPLC Glycan BEH Amide column (Waters, Milford, MA, USA). For the equilibration and binding 50 mM ammonium formate pH 4.4 was used as mobile phase, while the mobile phase for the elution was 100 % acetonitrile. Labeled N-glycans were detected with a fluorescence detector (ACQUITY UPLC FLR, Waters, Milford, MA, USA) at wavelength of E_x 265 nm/E_m 425 nm.

2.8 Enzyme linked immunosorbent assay for measurement of CHO host cell proteins

The CHO host cell protein content (HCP) of the bioreactor cell suspension and the elution samples was determined using a CHO HCP ELISA kit 3G (#F550) from Cygnus (Southport, NC, USA) following the manufacturer instructions. The HCP clearance was calculated as logarithmic removal factor (LRV) according to the following equation (Tomic et al. 2015):

$$LRV = \log \frac{ng \text{ HCP/mL cell broth}}{ng \text{ HCP/mL eluate}} \text{ (eq. 1)}$$

2.9 Statistical Analysis

Statistical comparisons were made using one-way analysis of variance and Student's unpaired t-test where $p < 0.05$ was considered significant.

3 Results

Processing high cell density culture suspension can be a challenging task in a manufacturing process. Here a novel integrated single-step is presented for a direct mAb capture from the cell suspension. The study includes three cell cultures (cult_01, cult_02 and cult_03) each followed by a single-step purification based on magnetic separation (run_01, run_02 and run_03). In all three runs the cells were grown in small scale bioreactors in which the medium was renewed using a UF HF-based cell separation, and magnetic bead purification was carried out on the non-clarified cell suspension immediately after culture harvest.

3.1 Intensified fed-batch cultures

Cultures cult_01, cult_02 and cult_03 were performed as described in Material and Methods. The evolution of the total cell density, viability and mAb concentration for these runs are given in Figure 1.

Cult_01 was a preliminary culture to test the technology and did not reach a very high cell density, i.e. maximum cell density of 23×10^6 cells/mL and cell density at harvest 17×10^6 cells/mL. The viability showed a rapid decline down to 84 % on the harvest day. On that day the UF HF filter clogged, provoking a volume increase to 350 mL and dilution of the culture, which was the reason for the apparent lower cell density at harvest. The filter failure was most probably due to the re-use of the filter. In cult_02, the cell density increased constantly from the inoculation density of 4.6×10^6 cells/mL to a maximum of 59.4×10^6 cells/mL, reached at day 10 and maintained at that level for the following days by cell bleeds until termination of the culture on day 16. At harvest day, the cell density was 56.8×10^6 cells/mL and the viability 86 %. In cult_03, the cell growth was comparable to cult_02 except that the inoculation density was lower, 1×10^6 cells/mL, another manufacturer was used for the filter and no cell bleed was applied. The density steadily increased, reaching 119×10^6 cells/mL on day 15, after which the UF HF filter clogged, causing a volume increase to 250 mL and a dilution of the cell concentration down to 103.9×10^6 cells/mL at day 16, i.e. the harvest day. During the whole run, the cell viability was higher than 95 %.

As presented in Figure 1B, the mAb accumulated in the bioreactor during all three runs. In cult_01, the concentration increased to 1.1 mg/mL at day 7, followed by a decrease to 0.8 mg/mL due to the higher harvest volume previously described. In cult_02, the final concentration was 6.6 mg/mL while it was 10.8 mg/mL in cult_03.

Notice that the apparent peak at day 13 in cult_03 was due to a temporary culture decrease at that day resulting in a transient higher mAb concentration.

3.2 Magnetic separation

The mAb capture process, i.e. run_01, run_02 and run_03, was initiated immediately after the bioreactor harvest of cult_01, cult_02 and cult_03 respectively. Figure 2A, 2C and 2E show the adsorption kinetics of the mAb from the cell suspension for the different runs monitored by the concentration of free mAb in samples taken at regular interval (see Material and Methods). In all the cases, the adsorption curves depicted an asymptotic behavior. A rapid mAb adsorption on the beads was observed within the first minutes with a steep decline of the free mAb concentration in the cell suspension. Looking at the proportion of adsorbed mAb in the 3 runs, the kinetics was fastest for the highest initial mAb concentration (run_03) and slowest for the lowest mAb concentration (run_01). In run_01, 91% of free mAb were adsorbed after 15 min, while this was achieved after 10 min for run_02 and an adsorption of 93 % of free mAb was obtained after 5 min for run_03. 97% of the mAbs were adsorbed after 30 min for run_01 and run_02, and after 10 minutes for run_03. In the latter, equilibrium was observed after 30 min, ≤ 60 min for run_02 and ≈ 60 min for run_01, for a final adsorption almost complete, 99% for run_01 and run_03, and 98% for run_02. These data showed a very rapid depletion of free mAb with high efficiency in a time much shorter than the total time of 2 hours used, for completeness of information, in this study.

Following the loading, the elution was performed in a semi-packed column to minimize the washing volume. This step had a total yield of 80 %, 90 % and 94% for run_01, run_02 and run_03 respectively (Figures 2B, 2D, 2F), i.e. the highest initial

mAb concentration generated the highest yield. The loss of mAb was comparable in run_02 and run_03, 0.66 g/L and 0.65 g/L respectively, and 0.16 g/L in run_01. The concentration factor obtained in the different runs was 3.9, 3.3 and 1.8 in run_01, run_2 and run_03, respectively.

3.3 mAb analysis

In order to assess the quality of the purified mAb and the properties of the purification method, the three culture and purification runs were analyzed with respect to the logarithmic clearance of HCP, distribution of monomer, high molecular weight fragments and low molecular weight fragments, the variations of mAb charge variants, and the distribution of glycosylation isoforms.

The reduction of the HCP is one of the most critical quality attributes for a purification process. The HCP levels in the culture harvests are listed in Table 1. Logarithmic HCP reductions of 2.9 for run_01 and 2.3 for run_02 and run_03 were achieved, generating a HCP concentration in the purification eluate of ≈ 0.6 ppm for run_01, ≈ 3.5 ppm for run_02, and ≈ 5 ppm for run_03.

The analysis of the mAb quality in the eluted material showed comparable profiles for all the runs (Figure 3), with low molecular weight (LMW) and high molecular weight (HMW) fractions below 4 %.

As shown in Figure 4, the charge variant analysis showed a distribution of ≈ 69 % main peak, ≈ 14.5 % low acidic and ≈ 16 % basic variants with a slight tendency for higher acidic variant and lower main peak levels for runs run_01, run_02 and run_03 in this order.

Finally, the analysis of the N-glycosylation pattern showed low levels of afucosylated glycan G0 and G1 in all the runs (Figure 5). The glycosylation maturation was higher in run_01 compared to run_02, in turns higher than run_03, as shown by a lower G2F and higher G0F contents in run_03 vs. run_01. In run_01 and run_03 high mannose was not detected, while it was very low in run_02.

4 Discussion

This study presents a downstream process based on magnetic bead separation for the capture of mAb directly from non-clarified high cell density cell suspension, illustrated in 3 runs. This process integrates the cell clarification with the mAb capture, enabling major operation simplification, particularly important for very high cell density cultures. In intensified fed-batch process as understood here, the culture medium is continuously changed via a UF-HF filter, generating accumulation of mAb but renewal of the components of lower molecular weight. These cultures were immediately followed by magnetic bead separation, where the antibodies were captured from the cell suspension without any pretreatment steps, before recovery by low pH elution, providing as well virus inactivation.

The high total yields of run_02 and run_03, 90 % and 94 %, demonstrated that the mAb capture from non-clarified high cell density cell suspension was possible in a very efficient way. In both runs, the free antibody depletion was very fast with more than 90 % of adsorbed mAb after 10 or 5 min. It was also very efficient as showed by 98 to 99 % adsorbed mAb at equilibrium. The high mAb titers of run_02 and run_03 were favorable to these high performances, which was expected since higher mAb concentrations promote higher diffusion rate and increase the probability of an antibody to be in the vicinity of a bead as illustrated by the lower total yield of 80 %,

and longer time to achieve 91 % of contact time of 15 min in run_01 compared to 10 and 5 min in run_02 and run_03. Comparable performances as run_01 had been observed previously with a yield of 86% for the purification of 1.51 mg mAb/mL from a fed-batch cell suspension (Brechmann et al. 2019).

A traditional protein A chromatography step performed on cell-free supernatant has typically a yield of 95 % (Kelley 2007). In combination with the cell clarification step, the overall yield of the clarification and capture steps is ≈ 90 % or less. In this perspective, the present technology has a higher yield for processes with high mAb titers.

Protein separation operated with a chromatography column (CC) or with magnetic beads in a stirred tank are, by essence, systems with different dynamics. A chromatography column is a tubular reactor or plug-flow reactor, while magnetic bead-based system is a batch stirred tank reactor (STR). The adsorption speed of mAb to the resin of a CC or to magnetic beads is similar at local level however the dynamics of the fluid is different with a faster mixing in the STR compared to the tubular reactor. Further, the residence time in traditional CC is a defining factor (Marichal-Gallardo and Álvarez 2012) that can be changed with the flow rate while the residence time for the magnetic bead STR is selected by the operator for the batch operation. For example, MabSelect SuRe LX (Cytiva, Uppsala, Sweden) provides a dynamic capacity of ≈ 40 mg IgG/mL resin at a residence time of ≈ 3 minutes, while a residence time of 30 minutes in the STR provides a binding of 97% of the antibodies to the magnetic beads. In the latter case, this time depends on the probability of an antibody molecule to be in vicinity of a bead and on the mixing time of the stirred tank; the diffusion speed in the porous bead is the same independently of the IgG or bead concentration. To assess the operation time for the CC and STR

systems, the steps of cell clarification, loading, washing and elution have to be taken into account. For the present conditions of 200 mL volume at 10.8 g IgG/L, a CC with a bead volume of ≈ 54 mL is needed to process the feed in one cycle, resulting in a loading time around 15 min. For the magnetic separation, a residence time of 30 minutes can be selected. The elution time is similar in both cases since it is based on packed bed. The cell clarification and wash steps add up a much longer operation time to the CC system compared to the magnetic separation. The clarification is already integrated in the magnetic separation system. Concerning the washing step, in a CC, working as a tubular reactor, the wash speed is similar to the loading time defined by the residence time. In the present example, a wash of 10 resin volumes (RV) takes ≈ 1.5 hour while in the magnetic separator a wash of 10 RV is realized within a matter of minutes.

These comparisons illustrate the excellent performances of magnetic bead separation performed on non-clarified culture harvest, in terms of yield and operation time, despite very high cell density $\geq 100 \times 10^6$ cells/mL.

To our knowledge the direct capture of mAb from mammalian cell suspension with a density larger than 100×10^6 cells/mL, has not previously been reported. The possibility to purify monoclonal antibodies from such a high cell density suspension has several advantages for manufacturing processes. The magnetic mAb capture is a single step replacing the clarification, by centrifugation and/or filtration, followed by the mAb capture. This reduces significantly the process time and highly enhances the productivity in comparison with legacy process including column chromatography (Brechmann et al. 2019; Ebeler et al. 2019; Hubbuch et al. 2001). Very high cell density cultures have a viscosity, which is much higher than the medium, e.g. twice the relative viscosity of medium for a cell density of 100×10^6 cells/mL and include a

large amount of cell debris (Clincke et al. 2013a). The clarification by centrifugation or filtration of very high cell density cultures is not possible nowadays. Filtration and centrifugation can reach their operating limits due to the high viscosity and high number of solid particles of a crude cell broth, resulting in additional filtration steps and additional pretreatments of the culture liquid such as preliminary dilution or flocculation, which introduce volume increase and/or supplementary step(s). The present approach shows the ability to overcome these problems for cell suspension at very high density. Another beneficial feature is that this process effectively copes with a high mAb titer, such as produced in process cult_03. Additionally, thanks to the removal of the clarification step, the associated step product loss is annihilated.

Protein aggregation is likely impairing the drug efficacy and can provoke an immune response (Dakshinamurthy et al. 2017). It is therefore essential to avoid aggregate formation or to remove them (Cromwell et al. 2006; Vázquez-Rey and Lang 2011). All three runs showed a comparable distribution of high molecular weight and low molecular weight fractions in the purified mAb with a monomer content of $\approx 95\%$ (Figure 3). This suggests that the magnetic bead separation was highly selective towards the monomer for all three runs. Given the capturing process of mAb based on protein A affinity for a free Fc region, a high selectivity for the monomer and/or fraction, displaying the free Fc region, can be expected (Hober et al. 2007). Comparing run_01, performed at moderate cell density, to runs run_02 and run_03, at high cell density, the high monomer content of $\approx 95\%$ did not differ greatly. This advocates that high cell density had no impact on the aggregate formation in the magnetic separation.

The mild mechanical conditions during the mAb capture using magnetic bead separation were also favorable to avoid the aggregation formation (Vázquez-Rey and Lang 2011; Wang et al. 2010).

The reduction of the host cell proteins (HCP) is one of the most important parameters for the purification process. All three runs provided a similar HCP reduction factor. The clearance in run_01 was slightly higher compared to run_02 and run_03, 2.9 vs. 2.3, and LRV values were in agreement with typical performances of protein A chromatography, i.e. $\approx 2.0 - 3.0$ LRV (Shukla et al. 2008; Xiao et al. 2020). Although the Health Authorities have no strict limit for HCP impurities in therapeutic proteins, it is common that the HCP level measured by ELISA is between 1 and 100 ppm after completion of the whole downstream process (Shukla et al. 2008). The present magnetic separation process reduced the HCP level down to values as low as 0.5 to 5 ppm after only one step. These very low levels indicated that the cells had been clarified during the magnetic separation in a very gentle way, which did not give rise to cell disruption and HCP increase. This represents a substantial advantage compared to the traditional cell clarification by centrifugation and/or filtration known to cause HCP increase from mechanical stress (Pohlscheidt et al. 2013).

Charge variations of mAb is a common phenomenon, which can be introduced by a variety of factors during the culture, intracellular and extracellular phenomena, during the purification or during the storage (Dakshinamurthy et al. 2017). Heterogeneity can be created either enzymatically or non-enzymatically and can include C-terminal lysine processing, N-terminal glutamine cyclization, deamidation, oxidation, formation of disulfide bonds, glycosylation, glycation and peptide bond cleavage (Dakshinamurthy et al. 2017; Liu et al. 2008). Those variations can have an impact on the efficacy, potency, immunogenicity and clearance from the patient

(Khawli et al. 2010; Meert et al. 2010; Neill et al. 2015). For Trastuzumab the receptor binding in two places can be affected by the heterogeneity. Firstly, the mAb binds specifically to the extracellular domain of the human epidermal growth factor receptor 2 (HER2) via the Fab region (Burstein 2005; Goldenberg 1999; McKeage and Perry 2002). The second important binding occurs between the Fc region of the mAb and the CD16 receptor on natural killer cells (NK) (Clynes et al. 2000; Dakshinamurthy et al. 2017). Those two binding effects are essential for the antibodies mechanism of action when used as a pharmaceutical for cancer treatment. The potential deamidation of asparagine residues into aspartate, at different positions of the CDR of purified Trastuzumab has been reported (Harris et al. 2001). The charge variant distribution observed in the presented study was very similar throughout the elution for all three purification runs (Figure 4). Compared to the charge variants in the originator Trastuzumab antibody (Dakshinamurthy et al. 2017), the present data showed a higher main peak percentage of 69.5 % vs. 56.7 % for the originator and slightly higher basic peak percentage of 16 % vs. 12.4 %. The acidic variants observed in the present study, in average 13.6 %, were notably lower compared to the originator 30.9 %. It is probable that the cultivation process influenced the charge variant distribution due to different conditions such as the cell line and culture medium. Furthermore, deamidation of light chain asparagine 30 in the complementary-determining region 1 (CDR1) and isomerization of heavy chain aspartate 102 in CDR3 are responsible for increased acidic variants in Trastuzumab (Du et al. 2012; Harris et al. 2001; Joshi and Rathore 2020). Both modifications have been shown to be the sites most susceptible to degradation in the CDR region of trastuzumab subjected to stress conditions of temperature and pH (Schmid et al. 2018), for which shifts can be introduced by storage and purification conditions.

Acidic variants are considered to be the main reason for the loss of potency of the binding towards the HER2 receptor (Dakshinamurthy et al. 2017) and a low percentage of acidic variants is desirable in order to achieve a high potency. On the contrary the most common basic variants are caused by lysine variations present in the C-terminal region of the heavy chain and play therefore no role in the binding activity towards the HER2 receptor. The presented study shows that the culture conditions together with the magnetic purification of mAb directly from cell broth have a beneficial effect on a lower formation of acidic variants.

The N-glycosylation is an important quality attribute, potentially influencing the protein stability, effector function and immunogenicity. The culture conditions have an important influence on the N-glycan distribution (Gomis-Fons et al. 2020; Zhang et al. 2019; Zhang et al. 2020). In the present study the N-glycosylation distribution showed a somewhat comparable profile in all three runs with however a tendency for a lower level of maturation for cult_03 compared to cult_02, and for cult_02 compared to cult_01. These differences were probably due to the culture conditions, which had not been optimized in this process; a CSPR of 20 pL/cell/day was applied in cult_02 and cult_03 vs. 50 pL/cell/day in cult_01. These variations were potentially within an acceptable range typically generated by culture process. In comparison with the glycosylation of originator and biosimilars reported in two recent publications, Joshi and Rathore 2020 and Dakshinamurthy et al. 2017 (see Supplementary Material), the present glycan distribution was in a comparable range. Notice however that afucosylated G1 glycan was observed in the present study while it seemed absent in the reference studies.

Finally, the analytical assessment of the eluate showed that the magnetic separation provided a high affinity for the mAb monomer, followed by highly

satisfying elution efficiency. The different evaluation performances were comparable for all the runs, indicating that the operation was independent of the cell density in the range studied here, i.e. between 23 and 104×10^6 cells/mL, and very efficient for all the studied mAb titers, i.e. up to 10.8 mg/mL.

We have previously reported that the magnetic separation process can easily be scaled up to pilot scale in case of a fed-batch process (Brechmann et al. 2019), while the present work addresses a much higher cell density. The main input parameters, which have to be selected for a magnetic separation, are the contact time of the cell suspension with the beads, i.e. binding step, the bead concentration (i.e. settled bead volume for the given sample volume), and the IgG load per bead volume. This latter is determined by the concentrations of IgG and beads. It could be at most the static binding capacity, however it should be lower in order to have a satisfying process time for the binding step. The influence of these parameters should be evaluated in small-scale, as presented in the present work, leading to the selection of values to be applied in the scaled-up process. A comparison of Figure 2C and Figure 2E provides an example on how the mAb concentration influences the binding kinetics. Both runs had a similar bead load of ≈ 42 mg mAb per mL of beads, which had been determined based on preliminary work. It was observed that run_03, which had the highest mAb and bead concentrations, showed the fastest binding of the free mAb in the cell suspension. The concentration of particles, i.e. beads and mAb, was higher in run_03, and this reduced the time necessary for the mAb binding to 15 min instead of 30 min in run_02. To get a faster binding kinetics, which represent saving in process time, a higher bead concentration can be used however this increases the cost of the resin. It can be interesting to decrease the binding step from 60 min to 30 min however below this time, the saving is not substantial.

As previously discussed, the magnetic separation system can be described by a STR, which means that the binding kinetic depends on the IgG adsorption on the beads and on the mixing in the STR. This mixing should be efficient but not harmful to the cells, i.e. similar to the mixing in the culture bioreactor. For the scale-up, the mixing time should be taken into account.

Furthermore, to design the process more economically it can be operated in multiple cycles reducing the total volume of beads similar to traditional CC, while keeping the ability to process large harvest quantities. This makes the magnetic separation for high cell density cell suspension a very interesting option from bench-top to large-scale purification. In view of the present study and previous experience of pilot scale magnetic separation, it is expected that the cell density will not influence the operation time.

5 Conclusion

The increasing demand of monoclonal antibodies places a growing burden on the manufacturing. The advances in upstream process, reaching very high cell densities and mAb titers, present a challenge for the downstream process. Very high cell density cultures contain a large amount of particles and have a much larger viscosity than medium, representing a major challenge for the primary recovery, i.e. centrifugation and filtration. Prior these, either flocculation, introducing a chemical agent, which has to be then purified, or dilution of the cell suspension are required (personal communication). This is sub-optimal, leads to prolonged process times, increases the expenses, causes product loss and exposes the mAb to harsh mechanical conditions, potentially resulting in aggregation formation or sometimes oxidation of methionine or tryptophan (Nowak et al. 2017). A streamlined, simpler and more

efficient process alternative to the steps of cell clarification followed by mAb capture can be therefore very advantageous, in particular for very high cell density cultures. The presented magnetic separation, integrating the cell elimination and magnetic bead-based capture, shows here excellent performances and can cope with the mentioned heavy burden placed on the downstream process. In the present study, it has been demonstrated that the very high concentrations of cell and particles in viscous cell suspension did not affect at all the mAb magnetic separation performances or the operation time. The present process had a high mAb adsorption, 98 to 99 %, and a high total yield, up to 94 %. The HCP reduction was comparable to protein A chromatography, 2-3 LRV, but, importantly, the cell elimination during the magnetic separation was very gentle so the final HCP was very low, between ≈ 0.6 and ≈ 5 ppm, this latter for a cell concentration of 100×10^6 cells/mL. The present study presents for the first time the direct capture of mAb from non-clarified cell suspension with a density larger than 100×10^6 cells/mL, and demonstrates that this process has excellent performances in terms of purity and yield. Previous experience of scaling up this type of process indicates as well that it is a very attractive alternative for manufacturing.

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7 Conflict of Interest

NB, KE and VC are co-inventors of a patent application related to the present study. KE is employee of MAGic Bioprocessing; the company that owns the intellectual property regarding the magnetic beads and the small-scale separators. HS, P-O E and AS declare no conflict of interest.

8 Author contributions

NB and VC designed the study. NB and HS performed the intensified fed-batch cultures. NB performed the magnetic bead-based purifications, process analytic, and result analysis. KE, P-OE, and AS gave scientific input on the cultivation and purification processes. NB and VC wrote the manuscript. NB, HS, KE, AS and VC revised the manuscript.

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10 Tables

Table 1: Process parameters for the mAb capture process performed with the harvest of cult_01, cult_02, and cult_03 using the MAGicBeads mAb beads and the MAGic SEP 500, and where the volume of settled beads was calculated according to the DBBC.

	<i>Run_01</i>	<i>Run_02</i>	<i>Run_03</i>
<i>Total cell density at culture harvest [total cells/mL]</i>	17.5×10^6	56.8×10^6	103.9×10^6
<i>Cell viability at harvest (%)</i>	84	86	96
<i>mAb titer at harvest [mg/mL]</i>	0.8	6.6	10.8
<i>HCP concentration in the harvest (ppm)</i>	507	630	1020
<i>Feed volume of the purification process [mL]</i>	300	200	200
<i>Volume of settled</i>	17.5	35	55.5

<i>magnetic beads [mL]</i>			
<i>Number of PBS washes</i>	3	5	5
<i>Time required for $\geq$ 98 % mAb adsorption (min)</i>	60	$\leq 60^{\dagger}$	45
<i>Total elution yield (%)</i>	80	90	94
<i>Concentration factor (v/v)</i>	3.9	3.3	1.8
<i>HCP concentration in the eluate (ppm)</i>	0.6	3.5	5
<i>HCP reduction</i>	3.0	2.3	2.3

[†] From the graph, a time shorter than 60 min is needed however an intermediate sampling would be required to identify more precisely this time

Figure legends

figure 1

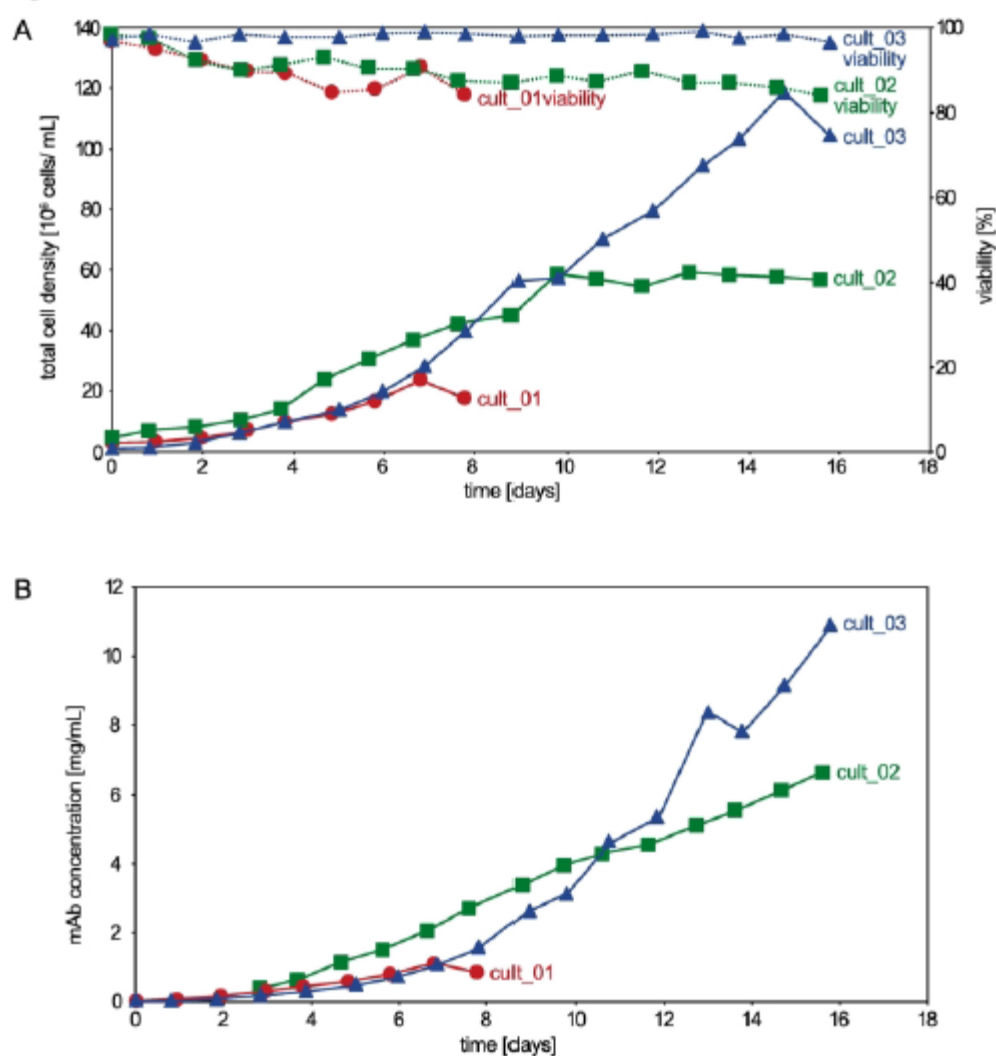


Figure 1 Cultivation data of the 3 intensified fed-batch cultures (cult_01, cult_02, cult_03). (A) Total cell density and viability as a function of cultivation time. (B) mAb concentration in the bioreactor as a function of cultivation time.

figure 2

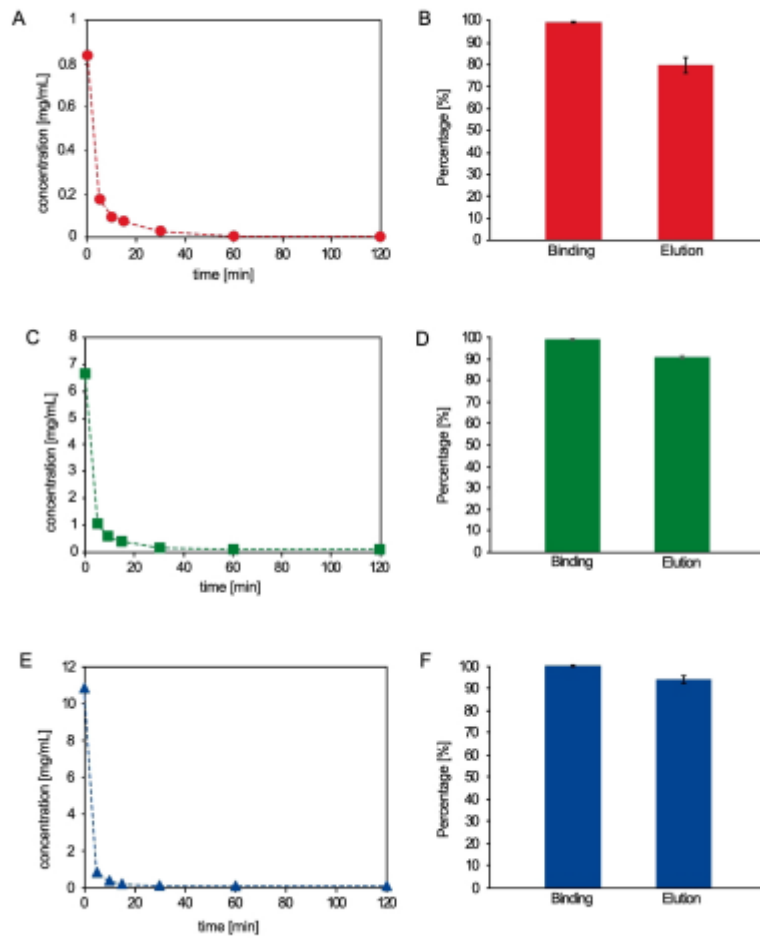


Figure 2 Magnetic purification directly applied to the harvested cell suspension; adsorption efficiency showed by the decrease with time of the free mAbs remaining in the cell suspension (left panels A, C, E); total binding after 2 h and total elution yield (right panels B, D, F). (A-B) Adsorption efficiency of run_01 with a mAb concentration of 0.8 g/L and a total cell density of 17.5×10^6 cells/mL. (C-D) Adsorption efficiency of run_02 with a mAb concentration of 6.6 g/L and a total cell density of 56.8×10^6 cells/mL. (E-F) Adsorption efficiency of run_03 with a mAb concentration of 10.9 g/L and a cell density of 103.9×10^6 cells/mL.

figure 3

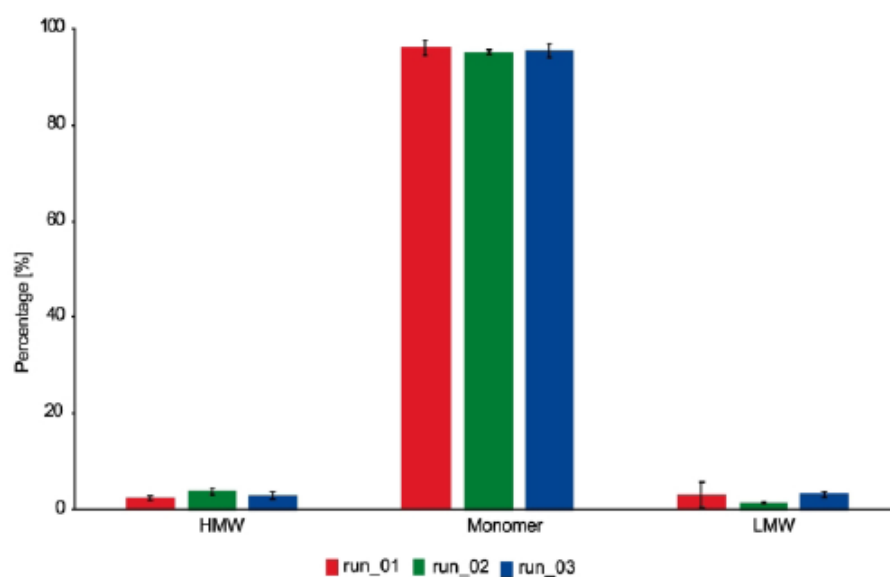


Figure 3 Size exclusion analysis showing the distribution of the monomers, high molecular weight (HMW) and low molecular weight (LMW) in the eluates of the three purification runs, run_01 (red), run_02 (green) and run_03 (blue) – error bar indicating the standard deviation of the analyses of 3 samples from separate elutions, where each analysis was the average of technical duplicate.

figure 4

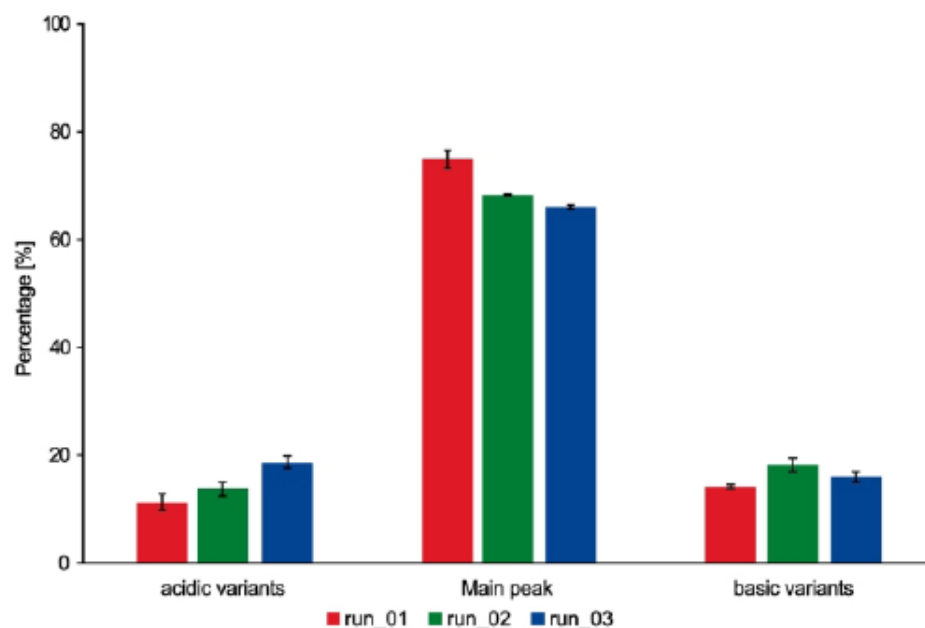


Figure 4 Charge variant distribution in the eluates of the three purification runs, run_01 (red), run_02 (green) and run_03 (blue) – error bar indicating the standard deviation of the analyses of 3 samples from separate elutions for run_01 and run_03 and 2 samples for run_02.

figure 5

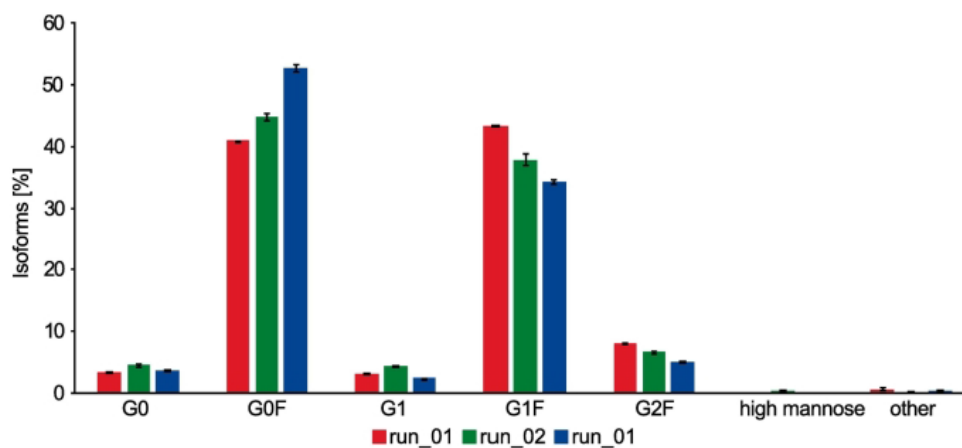


Figure 5 N-Glycosylation profiles of the eluates of the three purification runs, run_01 (red), run_02 (green) and run_03 (blue) – error bar indicating the standard deviation of the analyses of 3 samples from separate elutions for run_01 and run_03 and 2 samples for run_02, where each analysis was the average of technical triplicate

Supplementary Figure 1 N-Glycosylation profile of the eluate from run_01 (red), run_02 (green), and run_03 (blue) in relation to the profile of the originator mAb (dashed black/white and dashed purple/white) and their biosimilars (white and purple) as given in Joshi and Rathore 2020 and Dakshinamurthy et al. 2017.